3.6.5 Solving Problems Involving Quadratic Equations

Steps to solve a problem

Step 1: Convert the word problem to a quadratic equation form

Step 2: Solve the quadratic equation obtained in any one of the above three methods.

Step 3: Relate the mathematical solution obtained to the statement asked in the question.

Example 3.36 The product of Kumaran's age (in years) two years ago and his age four years from now is one more than twice his present age. What is his present age?

Solution Let the present age of Kumaran be x years.

Two years ago, his age $= (x - 2)$ years.

Four years from now, his age $= (x + 4)$ years.

Given, $(x-2)(x+4) = 1 + 2x$

$x^2 + 2x - 8 = 1 + 2x \Rightarrow (x-3)(x+3) = 0$ then, $x = \pm 3$

Therefore, $x = 3$ (Rejecting -3 as age cannot be negative)

Kumaran's present age is 3 years.

Example 3.37 A ladder 17 feet long is leaning against a wall. If the ladder, vertical wall and the floor from the bottom of the wall to the ladder form a right triangle, find the height of the wall where the top of the ladder meets if the distance between bottom of the wall to bottom of the ladder is 7 feet less than the height of the wall?

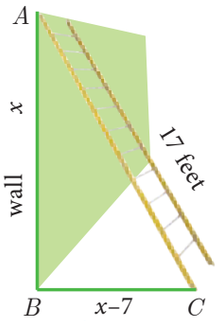


Fig. 3.7

Solution Let the height of the wall $AB = x$ feet

As per the given data $BC = (x–7)$ feet

In the right triangle ABC, $AC = 17$ ft, $BC = (x–7)$ feet

By Pythagoras theorem, $AC^2 = AB^2 + BC^2$

$(17)^2 = x^2 + (x-7)^2$; $289 = x^2 + x^2 - 14x + 49$

$x^2 - 7x - 120 = 0$ hence, $(x-15)(x+8) = 0$ then, $x = 15$ (or) -8

Therefore, height of the wall AB $= 15$ ft (Rejecting -8 as height cannot be negative)

Example 3.38 A flock of swans contained x^2 members. As the clouds gathered, $10x$ went to a lake and one-eighth of the members flew away to a garden. The remaining three pairs played about in the water. How many swans were there in total?

Solution As given there are x^2 swans.

As per the given data $x^2 - 10x - \frac{1}{8}x^2 = 6$ we get, $7x^2 - 80x - 48 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{80 \pm \sqrt{6400 - 4(7)(-48)}}{14} = \frac{80 \pm 88}{14}$$

3 Algebra.indd 115 16-12-2020 01:05:59 PM